

a.

Conditions: classes = (0, 1), n = 10000, hidden_layer_sizes = (9,), max_iter = 200
Results: Us: 0.998; Sklearn: 0.993 Logreg: 1.000

Conditions: classes = (0, 9), n = 9000, hidden_layer_sizes = (8,), max_iter = 180
Results: Us: 0.989; Sklearn: 0.983 Logreg: 0.997

Conditions: classes = (0, 3), n = 6500, hidden_layer_sizes = (7,), max_iter = 90
Results: Us: 0.996; Sklearn: 0.989 Logreg: 0.992

Conditions: classes = (0, 2), n = 2000, hidden_layer_sizes = (22,), max_iter = 130
Results: Us: 0.962; Sklearn: 0.923 Logreg: 0.987

Conditions: classes = (0, 4), n = 7000, hidden_layer_sizes = (49,), max_iter = 130
Results: Us: 1.000; Sklearn: 0.963 Logreg: 1.000

Conditions: classes = (0, 6), n = 1000, hidden_layer_sizes = (41,), max_iter = 180
Results: Us: 0.996; Sklearn: 1.000 Logreg: 0.993

Conditions: classes = (0, 4), n = 8705, hidden_layer_sizes = (1,), max_iter = 169
Results: Us: 0.988; Sklearn: 0.971 Logreg: 0.994

Conditions: classes = (0, 2), n = 20000, hidden_layer_sizes = (5,), max_iter = 188
Results: Us: 0.985; Sklearn: 0.978 Logreg: 0.989

b.

Conditions: hidden_layer_sizes(32,), max_iter = 100, sklearn_kwargs = {}, X, y = load_digits(classes=range(10), n=10000)
Results: Sklearn: 0.913; Logreg: 0.895

Conditions: hidden_layer_sizes(32,), max_iter = 100, sklearn_kwargs = {}, X, y = load_digits(classes=range(10), n=10000)
Results: Sklearn: 0.933; Logreg: 0.905

Conditions: hidden_layer_sizes(50,), max_iter = 50, sklearn_kwargs = {}, X, y = load_digits(classes=range(10), n=15000)
Results: Sklearn: 0.915; Logreg: 0.904

Conditions: hidden_layer_sizes(40,), max_iter = 40, sklearn_kwargs = {}, X, y = load_digits(classes=range(10), n=12300)
Results: Sklearn: 0.885; Logreg: 0.904

